GRAYCO
CONSTRUCTION
SERVICES

GRAYCO EMPLOYEE TRAINING

Fire & Smoke Restoration

GrayCo's own training for our crews. Covers the subjects of the IICRC FSRT (Fire and Smoke Restoration Technician) course, so it is also your study guide for that exam.

IN THIS COURSE

1. How GrayCo runs a fire job
2. What fire and smoke are
3. Safety on a fire job
4. Emergency response (the first visit)
5. Odor control on day one
6. How we clean
7. Cleaning the structure
8. Cleaning contents
9. Knowledge check and completion record

HOW TO USE THIS MANUAL

1. Read one module at a time.
2. Tick it off in the CRM under **Training**.
3. Take the test in the CRM: each answer shows right away.
4. Score **16 of 20** or better to pass.
5. Ask your supervisor about anything unclear.

TRAINEE

Name _____

Start date _____

Trainer _____

MODULE 1 OF 8

How GrayCo runs a fire job

KEY TERMS IN THIS MODULE

Insured | Insurance carrier | Pretest | Track the smoke | Restore or replace

On a fire loss we work for the **property owner (the insured)**. When the loss is an insurance claim, the **insurance carrier** is our second customer: they pay for the work, so every decision to clean or replace has to make sense to both of them.

- **Pretest before you promise.** Never tell a customer an item can be saved until you have tested it with the cleaning method you plan to use and talked it through with them.
- **Track the smoke.** Start at the room where the fire started and follow the residue in every direction it travelled until you find clean surfaces. Write down every material it touched.
- **Restore or replace?** That call is made only after the material is identified and pretested, and it is agreed with everyone who has an interest in the property.
- **Never name the insurance company to the customer** in writing; the CRM says "your insurance company" for you.

The most common mistake on fire jobs is skipping the pretest or starting work before the customer signs.

☐ I finished Module 1. Tick it off in the CRM: Forms & Docs → Training → GrayCo Fire & Smoke Restoration.

MODULE 2 OF 8

What fire and smoke are

KEY TERMS IN THIS MODULE

Combustion | Incomplete combustion | Microns | PAHs | Protein residue | Natural residue | Synthetic residue | Natural wicks | Heightened awareness odor

Combustion is fast, self-sustaining oxidation. It needs three things: **heat, fuel and oxygen**. Smoke and soot are the visible leftovers of **incomplete combustion**.

- Smoke particles are tiny: about **0.1 to 4 microns**. A micron is 1/25,000 of an inch, so you cannot see one particle with the naked eye.
- Fire residue falls into two groups: **PAHs** (polycyclic aromatic hydrocarbons, a gas-phase VOC) and **soot particles**. Both have been reported to be **carcinogenic**, so everyone on a fire site wears a fitted respirator with HEPA filters.
- **Less oxygen = more soot.** A smoldering, oxygen-starved fire makes heavy soot; a fire with plenty of oxygen burns cleaner.
- How far smoke travels through a building depends on the **air pressures and temperatures** the fire created.

The three residue types

- **Protein** – burned meat, fish or beans. Yellow to brown, greasy, almost invisible but a terrible odor. Needs thorough wet (water-based) cleaning.
- **Natural** – burned wood and paper. Gray to black, dry and powdery. Vacuum and dry-clean first, then wet clean. The easiest of the three.
- **Synthetic** – burned plastics, carpet, upholstery, drapes, toys. Thick, black and smeary. Don't touch it: vacuum it off first or you will smear it into the surface.

Other things to know

- **Natural wicks** are porous materials that soak up odor: fabric, unfinished wood, paper, unsealed concrete.
- Smoke carries vaporized oils that settle as an **oily film**, which is why residue can be hard to clean.
- Light-colored plastics hit by heat and smoke can turn **yellow**, and that can't be reversed economically.
- **Heightened awareness odor** (psychological odor) is when someone who lived through the fire keeps smelling smoke after it is gone. It is the hardest odor to overcome. Deodorizing early helps prevent it.

☐ I finished Module 2. Tick it off in the CRM: Forms & Docs → Training → GrayCo Fire & Smoke Restoration.

MODULE 3 OF 8

Safety on a fire job

KEY TERMS IN THIS MODULE

SDS | Labeled containers | Informed consent | HEPA air scrubber | Respirator | Lead (pre-1978) | Asbestos (pre-1977)

Safety is **everyone's** job. A burned building has hazards you can't see: weak floors and roofs, live wiring, falling debris, sharp edges and slick surfaces.

- GrayCo issues the **safety equipment** and the **safety training** you need (OSHA requires both). Use them.
- You learn what a chemical can do from three places: the **SDS** (safety data sheet), the **label**, and **training** from the maker or supplier.
- Every chemical stays in a **properly labeled** container and is secured on site.
- **Chemically sensitive customers** should not be **in or near** the work areas until we are done. Get their **informed consent** before using any deodorant. They may need extra air scrubbing and gentler products.
- Where there is loose soot, run **HEPA air scrubbers** and exhaust fans. **PPE** is the last line of defense after those engineering controls. The most important piece is a **respirator with HEPA filtration**.

Lead and asbestos

- **Lead** is a heavy metal found in paint made before 1978. **Asbestos** was used as insulation and fireproofing in siding, roofing, flooring and adhesives before 1977.
- Both get into the air mostly during **sanding and demolition**.
- If you think a material has lead or asbestos, **stop removing it** and have it tested. Follow GrayCo's lead standard: shut the HVAC off, post the no eating, drinking or smoking signs (English and Spanish), and the PM opens a change order. Lead, asbestos and mold can cause serious lung, nerve and kidney damage, so don't take shortcuts.

☐ I finished Module 3. Tick it off in the CRM: Forms & Docs → Training → GrayCo Fire & Smoke Restoration.

MODULE 4 OF 8

Emergency response (the first visit)

KEY TERMS IN THIS MODULE

Documentation | Work authorization | Structural integrity | Cold surfaces | Board-up | Acidic soot | Corrosion | Loss mitigation

The most important thing you bring to a fire job is the right **attitude**. The family has just lost a lot; be calm, kind and clear.

- **Document everything first.** Photos and notes of every room and its condition before you touch anything. Customers who watch you document are much less likely to blame us later for damage that was already there.
- **Work authorization.** It is a binding agreement with whoever is financially responsible, usually the **insured or property owner**. No work starts until it is signed.
- **Walk the building carefully.** Watch for hazards and the building's **structural integrity**. Wear your PPE.
- **Where smoke ends up:** hot smoky air is drawn to **cold surfaces** (outside walls, windows, metal), horizontal surfaces, and plastics that carry static. Check closets, wall cavities, attics and basements.
- **Board-up** must give **durable protection** and cause **minimal additional damage** to the building. Options are cover-over, fitted inserts, bolt tension, tarps and shrink-wrap.
- Soot mixed with moisture in the air turns **acidic** and starts to stain and corrode. Neutralizing it, usually with alkaline cleaners, is the first step in **corrosion** prevention.
- **Loss mitigation** work stops more damage: winterizing so pipes don't freeze, controlling corrosion, and cleaning surfaces before they stain.

☐ I finished Module 4. Tick it off in the CRM: Forms & Docs → Training → GrayCo Fire & Smoke Restoration.

MODULE 5 OF 8

Odor control on day one

KEY TERMS IN THIS MODULE

Four principles | Pairing agent | Masking agent | ULV fogger | Thermal fogger | Oxidizers | Ozone | Hydroxyl | Pack-out

Odors are strongest when **humidity and temperature are elevated**, which is exactly how a building is right after the fire is out.

- Pull out **charred, unsalvageable** materials early. It cuts **odors and further soiling**.
- Heavy smoke odor always takes a **combination** of methods, never one.
- Deodorize **early**: it shows the customer progress, gives the crew a better place to work, and helps prevent heightened awareness odor.
- **Start in the room where the fire began** and work out from there.

Four principles of odor control

1. **Remove the source** (if you can).
2. **Clean the source area.**
3. **Penetrate**: get the deodorant everywhere the smoke went.
4. **Seal** the surfaces that still hold odor.

Products and equipment

- **Pairing agents** combine with the odor molecule to make a new compound with no smell. **Masking agents** cover an odor with a stronger, pleasant scent. Scented oils, gels, solids and granules are masking agents.
- **ULV (ultra low volume) foggers** make 8–15 micron droplets of water-based deodorant that stay airborne for hours.
- **Thermal foggers** use heat to make a dry, very fine (about 0.5 micron) fog that travels like smoke.
- **Oxidizers**: **chlorine bleach** (liquid; it discolors nylon, dissolves wool and silk and corrodes some metals), **ozone** (O3, a gas made by an ozone machine) and **hydroxyl radicals** (made from **ultraviolet** light and water vapor). Ozone can hurt people, pets, plants and natural rubber.
- Measure every product and follow the **label instructions**. Ask the customer before adding fragrances.

Before you fog

- Call the **fire department, police or alarm company** before **thermal fogging**, so nobody reports a fire.
- Too much petroleum fog in a tight space can cause a **possible explosion, breathing irritation and slippery residue**.

- Anyone without the right PPE must **vacate** areas being wet fogged, thermal fogged or ozone treated.

Pack-out

- Food in boxes or cans that got hot is **documented**, then thrown away.
- Precondition everything you pack out with a **lamb's wool duster or a vacuum with a brush attachment**.

☐ I finished Module 5. Tick it off in the CRM: Forms & Docs → Training → GrayCo Fire & Smoke Restoration.

MODULE 6 OF 8

How we clean

KEY TERMS IN THIS MODULE

Preconditioning | Washable | Non-washable | Alkaline cleaner | Water first | Wood restoring cream

- **Preconditioning** removes loose soot dry, before any liquid touches it: lamb's wool duster, HEPA vacuum with a soft brush, compressed air, or a dry sponge.
- **Washable** surfaces can take water-based cleaning. **Non-washable** surfaces can't.
- Most smoke residue is **acidic**, so most of our cleaners are **alkaline**.
- **Mixing rule:** put the **water** in the container first, then add the **concentrate**. You get the right mix and no foam.
- **Wood restoring cream** is for finished wood with heavy smoke. Always work with the grain.
- Good cleaning comes from six things working together: the chemical, its reaction, lubrication, agitation, temperature and dwell time.

☐ I finished Module 6. Tick it off in the CRM: Forms & Docs → Training → GrayCo Fire & Smoke Restoration.

MODULE 7 OF 8

Cleaning the structure

KEY TERMS IN THIS MODULE

Heat line | Ceiling to floor | Bottom to top | Dry sponge | Sealer | Media blasting | Etching | Puff-back | NADCA

Cleaning is usually the cheapest way to restore. Our job is to **preserve fabrics and surfaces** from further damage. In light-to-moderate jobs, start by cleaning the **source area**.

- The **heat line** is the visible line where it got very hot. Above it, damage is heavier; it tells you how restorable a room or a piece of furniture is.
- Popcorn (acoustic textured) ceilings **deteriorate** if you clean them hard. Plan to clean for paint or sealer only.
- **Secure breakables** and take down drapes and pictures before wet cleaning.
- Work **ceiling to floor**. The exception is washing painted walls by hand: go **bottom to top** so drips don't streak.
- Gloss and semi-gloss paint: a **general purpose cleaner or degreaser**. Flat latex: **precondition or dry sponge first**; if it has to be wet cleaned, plan to repaint.
- If stains or odor remain, **seal** before painting (a stain-blocking sealer).
- Attics are hard because of stored items of **questionable value**, rough surfaces, and tight spaces.
- Charred framing: scrape and seal, **blast** it (abrasive or wire brush), or remove and replace.
- Finished wood paneling: wipe with the **wood grain** and wet clean with a **mild alkaline detergent**, then polish.
- Masonry: let it **dry** before you judge it. Use **alkaline** cleaners and **media blasting**; if those fail, **acid** cleaners, which **etch** the surface. Masonry floors may have a **sealer or finish** on top.
- Vinyl wall covering: a **moderate** alkaline cleaner. Grass cloth is the hardest wall covering to treat.
- Salvageable floors get cleaned at the start **and again at the end of the job**.
- **Fiberglass** tubs and showers stain the easiest; use non-abrasive **cream, foam or liquid** soft-scrub. Grout takes **heavy-duty alkaline or mild acid** cleaners. Acid cleaners also work on tile and **aluminum window frames**. In small rooms, **ventilate and wear PPE**.
- Acoustical ceiling tile (no asbestos) with light soot: **vacuum** first, then dry sponge.
- Light fixtures: **turn off and tag out** the power first.
- Kitchens: clean the **kitchen ventilation system**, and don't miss the hood filters, **fans and exhaust ducts**. On built-in appliances, clean the **mounting cavities and subsurface areas**.
- A **puff-back** is an oil furnace misfire that blows soot through the ducts. An **inspection** tells you whether the whole HVAC system needs cleaning; heavy soot is cleaned to **NADCA standards** (National Air Duct Cleaners Association).

- [ ] I finished Module 7. Tick it off in the CRM: Forms & Docs → Training → GrayCo Fire & Smoke Restoration.

MODULE 8 OF 8

Cleaning contents

KEY TERMS IN THIS MODULE

Three values | 0000 steel wool | Dry soil removal | Dry solvent | Ticking | Ultrasonic | Pad and strap

Weigh three kinds of value before spending time on an item: **monetary, antique and sentimental**. Anything we can't save is **documented** and the customer approves before it is thrown out.

Wood furniture

- Check where the piece sat compared with the **heat line**.
- Use **0000** steel wool, the finest grade, with wood restoring cream for baked-on soot and white rings.
- Pieces with a **clear finish** respond best to wood restoring cream. White haze is a reaction to **heat or moisture**.
- Finish with **furniture polish**. Don't **stack** furniture right after cream; it can mark the finish.
- Blistered or badly stained pieces go to a **refinisher** or get replaced. **Light sanding** can take small stains out of unfinished wood.
- Petroleum deodorants can **darken** porous wood; never use them without the customer's OK.

Fabrics

- Before cleaning upholstery, check **fiber and weave, existing conditions, and how much and what kind of soot**.
- Step one on any fabric is **dry soil removal**. Use **dry solvent** where water could shrink, brown or bleed it. Newer, lightly sooted upholstery can be done by dry vacuuming, air washing or solvent cleaning.
- Clothing is cleaned and deodorized **as soon as practical**. Have the customer **select** what to keep and what to toss. Wire and plastic hangers get **thrown out right away**.
- Mattress cover fabric is called **ticking**, and it is usually colorfast. Stains that were already there won't come out.
- Drapes: note uneven hems, weak fabric and water marks before you start.
- Blinds: clean on a **protected surface** or in an **ultrasonic** tank.
- Lampshades: watch for **colored trim** that can bleed, and handle them only by the **metal frame**.

Other contents

- **Electronics**: inspected first, then cleaned and tested by a qualified tech.
- **Toys**: inspect, clean, sanitize.
- **Art**: oil and acrylic paintings restore best. **Document** the condition of anything valuable.

- Personal papers are **turned over** to the customer.
- **Books**: check what they're made of, any moisture, and how much heat and smoke damage. Dry clean with a dry sponge, crumbly dough cleaner or art gum eraser.
- **Dishes**: clean as soon as practical; presoak heavy soot in a **degreaser**; wrap in white **newsprint or bubble wrap**. Fine china and crystal: disclaim hidden damage, and wash **by hand** only.
- **Moving and storage**: never let pieces touch surface to surface. **Pad and strap** everything before loading, and **protect** stored contents from recontamination.

☐ I finished Module 8. Tick it off in the CRM: Forms & Docs → Training → GrayCo Fire & Smoke Restoration.

KNOWLEDGE CHECK

Test yourself

Fill in the bubble for the best answer. This is the same test as in the CRM, where each answer shows right away and your score is saved. You need **16 of 20** to pass.

1. **Who is GrayCo's primary customer on a fire loss?**
   - (A) The insurance adjuster
   - (B) The property owner (insured)
   - (C) The mortgage company
   - (D) The fire department

2. **Before telling a customer an item can be restored, you must:**
   - (A) Get the adjuster's approval
   - (B) Pretest it and talk with the customer
   - (C) Pack it out
   - (D) Deodorize it

3. **Smoke and soot are by-products of:**
   - (A) Complete combustion
   - (B) Oxidation of metals
   - (C) Incomplete combustion
   - (D) Evaporation

4. **About how big are smoke particles?**
   - (A) 0.1 to 4 microns
   - (B) 25 to 100 microns
   - (C) 1/8 inch
   - (D) 100 to 500 microns

5. **Burned plastics and synthetics leave a residue that is:**
   - (A) Dry and powdery
   - (B) Yellow and greasy
   - (C) Thick, black and smeary
   - (D) Invisible

6. **Burned meat, fish or beans leave which residue type?**
   - (A) Natural
   - (B) Protein
   - (C) Synthetic
   - (D) Mineral

7. **Soot contamination is heaviest when the fire had:**

   (A) Plenty of oxygen
   (B) A lack of oxygen
   (C) Sprinklers
   (D) Low heat

8. **Porous materials that hold odor (fabric, paper, unfinished wood) are called:**

   (A) Natural wicks
   (B) Heat lines
   (C) Masking agents
   (D) Smoke webs

9. **The most important PPE where toxic particles may be present is:**

   (A) Hard hat
   (B) Respirator with HEPA filtration
   (C) Safety glasses
   (D) Steel-toe boots

10. **Lead and asbestos are most likely released during:**

    (A) Painting and caulking
    (B) Sanding and demolition
    (C) Vacuuming
    (D) Deodorizing

11. **Who signs the work authorization?**

    (A) The adjuster
    (B) The tenant
    (C) The insured or property owner
    (D) The fire marshal

12. **Soot combined with water vapor in the air becomes:**

    (A) Alkaline
    (B) Neutral
    (C) Acidic
    (D) Flammable

13. **Odors are most noticeable when humidity and temperature are:**

    (A) Elevated
    (B) Low
    (C) Freezing
    (D) It makes no difference

14. Which is NOT one of the four principles of odor control?

- (A) Remove the source
- (B) Clean the source area
- (C) Mask with fragrance
- (D) Seal surfaces

15. Before thermal fogging you should:

- (A) Turn up the heat
- (B) Notify the fire department / authorities
- (C) Open the windows only
- (D) Spray water first

16. When mixing cleaning chemicals:

- (A) Put concentrate in first
- (B) Put water in first, then concentrate
- (C) Mix in hot water only
- (D) Shake before adding water

17. Painted walls are washed by hand in which direction?

- (A) Top to bottom
- (B) Bottom to top
- (C) Left to right
- (D) Middle out

18. Acid cleaners remove a very thin layer of masonry. This is called:

- (A) Etching
- (B) Blushing
- (C) Pitting
- (D) Sealing

19. The bath fixture surface most easily stained by smoke is:

- (A) Porcelain
- (B) Cast iron
- (C) Fiberglass
- (D) Ceramic tile

20. Fine china and crystal with heavy smoke should be cleaned:

- (A) In a dishwasher
- (B) By hand
- (C) With ozone
- (D) With bleach

COMPLETION RECORD

You finished the course

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CERTIFICATE OF COMPLETION

GrayCo Fire & Smoke Restoration

GrayCo Construction Services, LLC employee training

Trainee name

Test score (out of 20)

Trainee signature

Trainer / supervisor

Date completed

Date reviewed

Keep a copy in the employee's file. The CRM keeps each person's test score under Training → Team progress.